

Dietary milled flaxseed and flaxseed oil improve N-3 fatty acid status and do not affect glycemic control in individuals with well-controlled type ...

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OBJECTIVE: To determine the effects of dietary consumption of milled flaxseed or flaxseed oil on glycemic control, n-3 fatty acid status, anthropometrics, and adipokines in individuals with type 2 diabetes. DESIGN: Thirty-four participants were randomized into a parallel, controlled trial. SUBJECTS: The participants were adults with type 2 diabetes (age 52.4 +/- 1.5 years, body mass index 32.4 +/- 1.0 kg/m(2), n = 17 men and 17 women). INTERVENTIONS: Participants consumed a selection of bakery products containing no flax (control group [CTL], n = 9), milled flaxseed (FXS, n = 13; 32 g/d), or flaxseed oil (FXO, n = 12; 13 g/d) daily for 12 weeks. The FXS and FXO groups received equivalent amounts of alpha-linolenic acid (ALA; 7.4 g/day). MEASURES OF OUTCOME: The primary outcome measures were fasting plasma hemoglobin A(1c), glucose, insulin, and phospholipid fatty acid composition. The secondary outcome measures were fasting circulating leptin and adiponectin, as well as body weight, body mass index, and waist circumference. Dietary intake assessment and calculations for homeostasis model assessment for insulin resistance and quantified insulin sensitivity check were also completed. RESULTS: The FXS and FXO groups had increases in plasma phospholipid n-3 fatty acids (ALA, eicosapentaenoic acid [EPA], or decosapentaenoic acid [DPA], but not docosahexaenoic acid), and the FXO group had more EPA and DPA in plasma phospholipids compared to the FXS group. All groups had similar caloric intakes; however, the CTL group experienced a 4% weight gain compared to baseline (p < 0.05), while both flax groups had constant body weights during the study period. All other parameters, including glycemic control, were unchanged by dietary treatment. CONCLUSIONS: Milled FXS and FXO intake does not affect glycemic control in adults with well-controlled type 2 diabetes. Possible prevention of weight gain by flax consumption warrants further investigation.